$A = \{1, 3, 5, 7, 12\}$ $B = \{4, 6, 60, 15\}$

$R = \{a/b\}$

$aRb \Leftrightarrow a/b$

$1R4$ $1R6$ ...

$3R6$ $3R60$ $3R15$

$5R60$ $5R15$

$12R60$

Reflexiva:

$aRa, \forall a \in A$

$A = \{1, 3, 5, 7\}$ $\quad 1R1, 3R3, 5R5, 7R7$

Simétrica:

$\forall a, b \quad aRb \Leftrightarrow bRa$

$2R6 \not\Rightarrow 6R2$ No simétrica

$R: \{a, b \in \mathbb{R} \mid b$ tiene la misma paridad $\}$

$aRb \Leftrightarrow bRa$ simétrica

antisimétrica

$$\forall a,b \in \mathbb{R} \quad aRb \text{ y } bRa \Rightarrow a = b$$

Divide a

$$\left.\begin{array}{l} a/b \\ b/a \end{array}\right\} \Rightarrow a = b$$

transitiva:

$$\left.\begin{array}{l} aRb \\ bRc \end{array}\right\} \Rightarrow aRc$$

$a/b \Rightarrow a \cdot n_1 = b \quad , n_1 \in \mathbb{N}$

$b/c \Rightarrow b \cdot n_2 = c \quad , n_2 \in \mathbb{N}$

$n_1 \cdot n_2 = n^* \in \mathbb{N}$

$b \cdot n_2 = a \cdot n_1 n_2 = c \Rightarrow$

$\Rightarrow a\, n^* = c \Rightarrow$

$\Rightarrow a \mid c$

transitiva

$A = \{1, 2, 3, 4\}$

$R = \{(a,b) \mid a, b \in A$ y a divide $b\}$ $\rightarrow an = b,\ n \in \mathbb{N}$

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

Reflexiva

$aRa \Rightarrow an = a$ ✓ $\boxed{n = 1} \in \mathbb{N}$

$n \in \mathbb{N}$

Simetría $\rightarrow$ $\boxed{a = b}$

$aRb \Rightarrow an_1 = b$

$bRa \Rightarrow bn_2 = a$

$\Rightarrow bn_1 n_2 = b \Rightarrow$

$\Rightarrow n_1 n_2 = 1 \rightarrow$

$\Rightarrow n_1 = 1$

$n_2 = 1$

$\boxed{\text{ANTISIMÉTRICA}}$

$aRb \Rightarrow an_1 = b$ (divide a)

$bRc \Rightarrow bn_2 = c$

$\Rightarrow an_1 n_2 = c \Rightarrow$

$\Rightarrow an = c \Rightarrow$

$n \in \mathbb{N}$

$\Rightarrow aRc$ transitive

Relaciones de equivalencia

Reflexiva, Simétrica, transitiva

clase de equivalencia

$$[a]_R = \{ x \in A \mid aRx \}$$

$$[1] = \{1, 2, 3, 4\}$$

$$[2] = \{2, 4\}$$

$$[3] = \{3\}$$

Relación congruencia

$\{(a,b) \mid a \equiv b \pmod{m}\}$

$m > 1, \; m \in \mathbb{Z}$

$8 \mid \frac{3}{2}$

2

a mod m

$3 \cdot n + x = 8$

$3 \cdot 2 + 2 = 8$

$\boxed{8 \bmod 3 = 2}$

$7 \bmod 2 = 1$

$11 \bmod 9 = 2$

$15 \bmod 4 = 3$

$\boxed{11 \bmod 3 = 2}$

$8 \equiv 11 \bmod 3$

$$\text{impar}_1 \equiv \text{impar}_2 \pmod 2$$

$7 \bmod 2 = 1$

$13 \bmod 2 = 1$

$147451479975 \bmod 2 = 1$

$17 \bmod 3 = 2 \longrightarrow 3 \cdot 5 + 2 = 17$

$23 \bmod 3 = 2 \longrightarrow 3 \cdot 7 + 2 = 23$

$$17 \equiv 23 \pmod 3$$

$\boxed{m = 3}$

$[2] = \{\ldots -4, -1, 2, 5, 8, 11 \ldots\}$

$\{x \in \mathbb{Z} \,/\, x \equiv 2 \mod 3\} = \ldots -4, -1, 2, 5, 8, 11 \ldots$

$3 \cdot n + 2 =$

$x \equiv 2 \mod 3$

$2 \mod 3$

$3 \cdot n + 2 = 2, 5, 8, 11$

-1

$2 \,\underline{|\, 3}$

$\quad\;\; 0$

2

Relación orden

Reflexiva, antisimétrica, transitiva

$aRb \Leftrightarrow a \leq b$

Reflexiva

$a \leq a$ ✓

Simetría

$aRb \Rightarrow a \leq b$

$bRa \Rightarrow b \leq a$

$\boxed{a = b}$

antisimétrica

transitiva

$a \leq b$

$b \leq c$ $\Rightarrow$

$\Rightarrow a \leq b \leq c \Rightarrow$

$\Rightarrow \boxed{a \leq c}$

Ser múltiplo de: $(\mathbb{N})$

$$aRb \Rightarrow b \cdot n = a$$

$aRa \Rightarrow a \cdot n = a \quad , n \in 1$ ✓

$aRb \Rightarrow bn_1 = a$

$bRa \Rightarrow an_2 = b$ $\quad \downarrow an_2 n_1 = a \Rightarrow$

$\Rightarrow n_2 = 1$

$n_1 = 1$

$bn_1 = a \Rightarrow b = a$

$\boxed{a = b}$ ✓

$aRb \rightarrow bn_1 = a$

$bRc \rightarrow cn_2 = b$

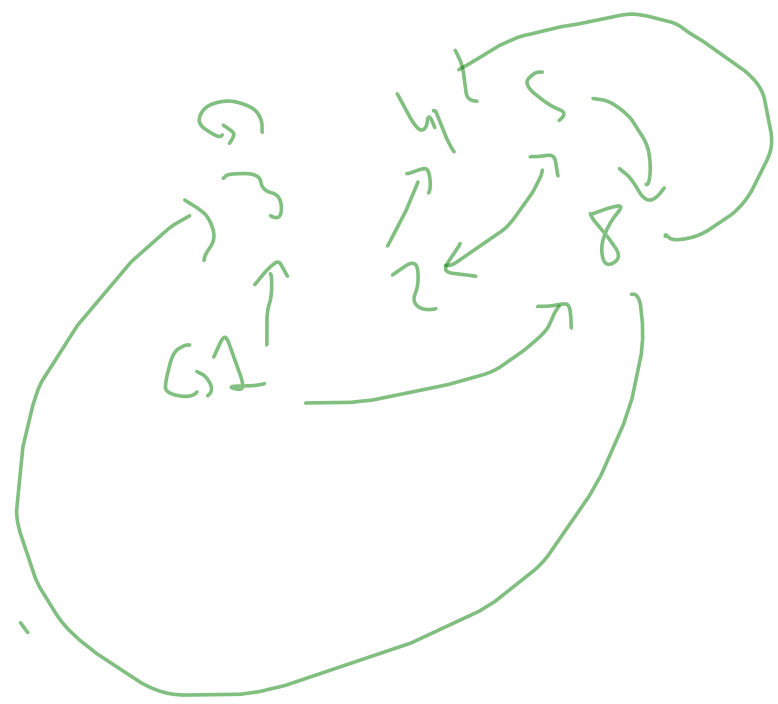
$cn_2 n_1 = a$

$cn = a \rightarrow aRc$

$\boxed{\text{Ordyn}}$

$\cancel{(1,1)}$, $\cancel{(1,3)}$, $\cancel{(1,8)}$, $\cancel{(2,5)}$, $\cancel{(2,4)}$, $\cancel{(3,1)}$

$\cancel{(5,8)}$, $\cancel{(5,2)}$, $\cancel{(8,4)}$, $\cancel{(8,3)}$



1 R 8 $\mapsto$ 1 R 3

8 R 3

2 R 5 $\mapsto$ $\cancel{2 R 8}$

5 R 8

Divisores 12

$A = \{1, 2, 3, 4, 6, 12\}$

$R = \{aRb \Leftrightarrow a|b\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,12),$
$(2,2), (2,4), (2,6), (2,12),$
$(3,3), (3,6), (3,12)$
$(4,4), (4,12)$
$(6,6), (6,12)$
$(12,12)\}$

$aRa \Leftrightarrow a|a \Rightarrow a \cdot 1 = a$

$aRb \Rightarrow a|b \Rightarrow a \cdot n_1 = b$

$bRa \Rightarrow b|a \Rightarrow b \cdot n_2 = a$ $\Big\} \Rightarrow$

$\Rightarrow n_2 \cdot n_1 = b \Rightarrow$

$a \cdot 1 = b \Rightarrow \boxed{a = b}$ $\Rightarrow$ $n_2 = 1$, $n_1 = 1$

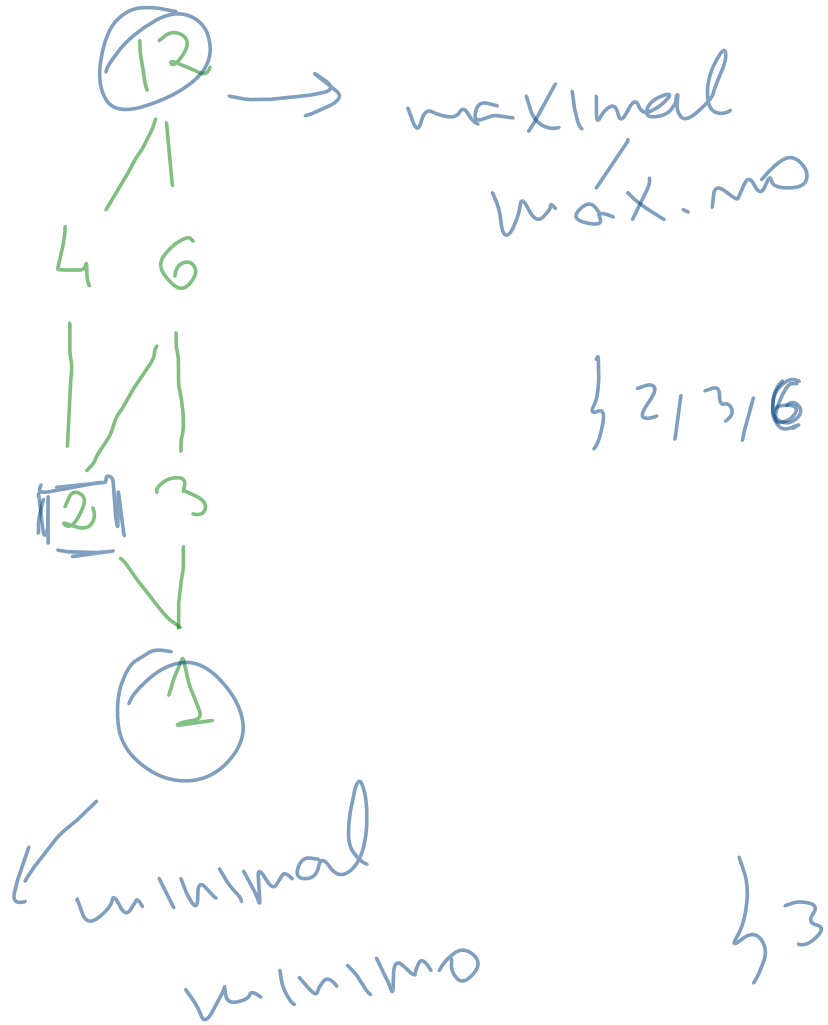
$aRb \Rightarrow a n_1 = b$

$bRc \Rightarrow b n_2 = c$ $\Big\} a n_1 n_2 = c \Rightarrow$

$\Rightarrow a n = c \Rightarrow$

$\Rightarrow a|c \Rightarrow$

$\Rightarrow aRc$



$\{2,3,6\} \to 12$ supremo

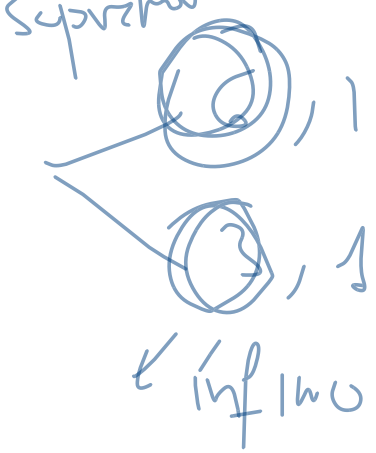
superior

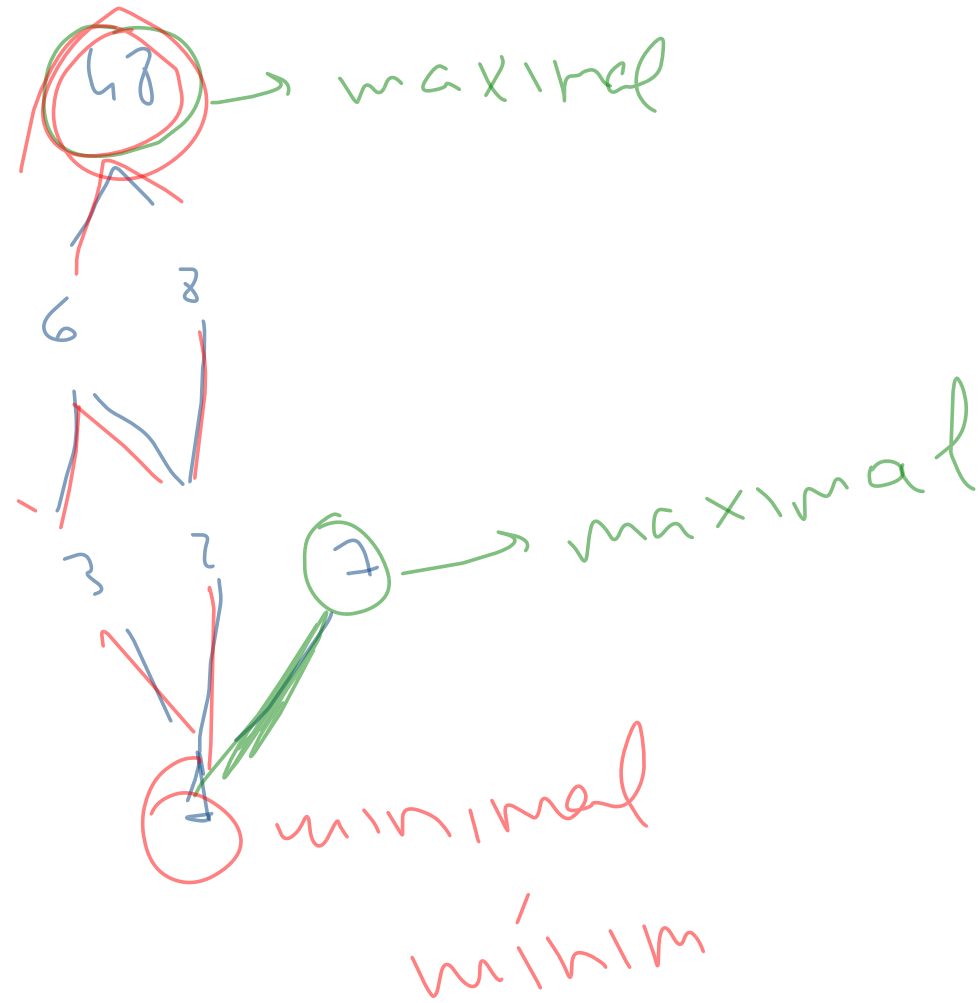
1 inferior

ínfimo

$\{3,6\}$

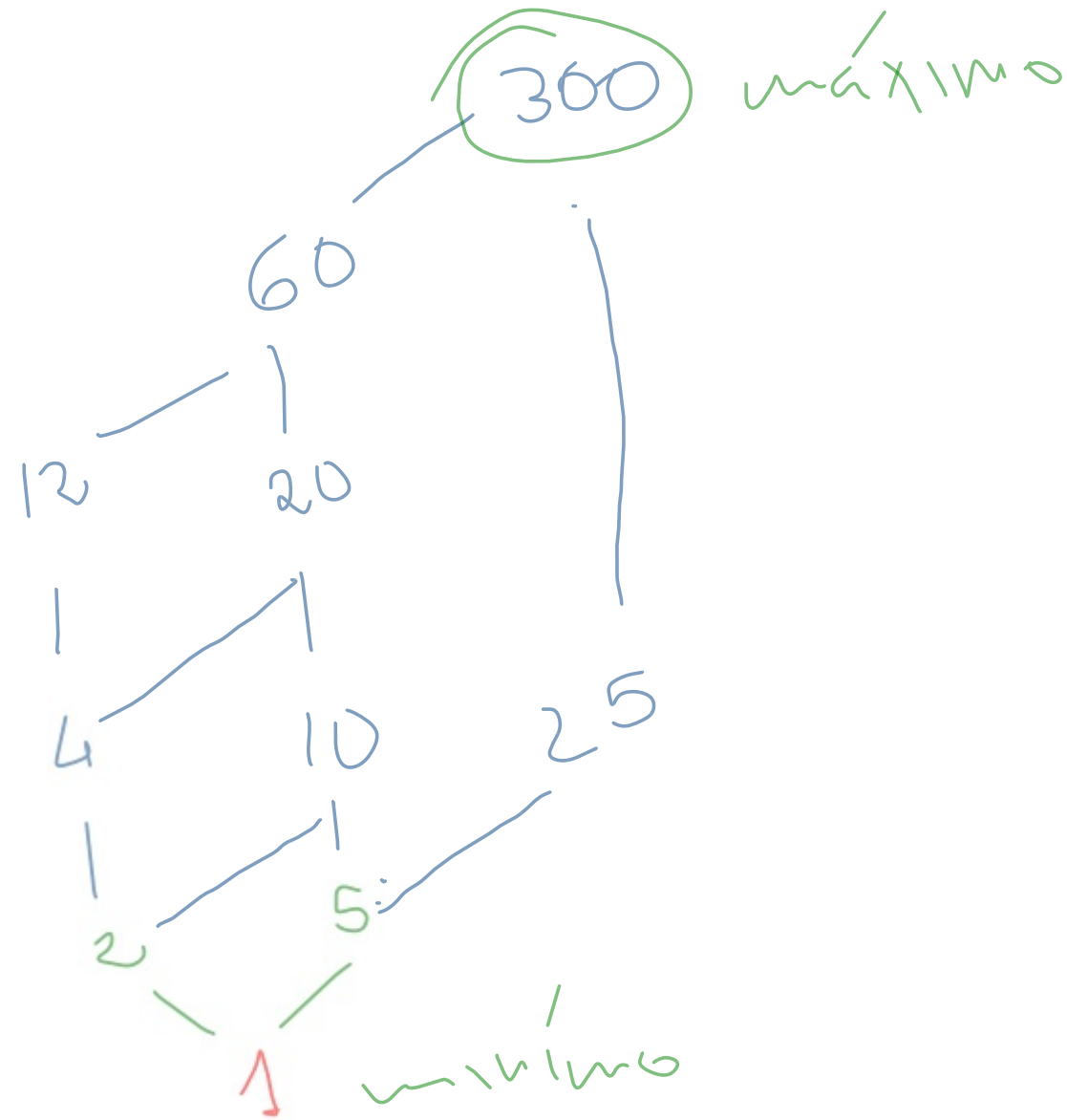
supremo



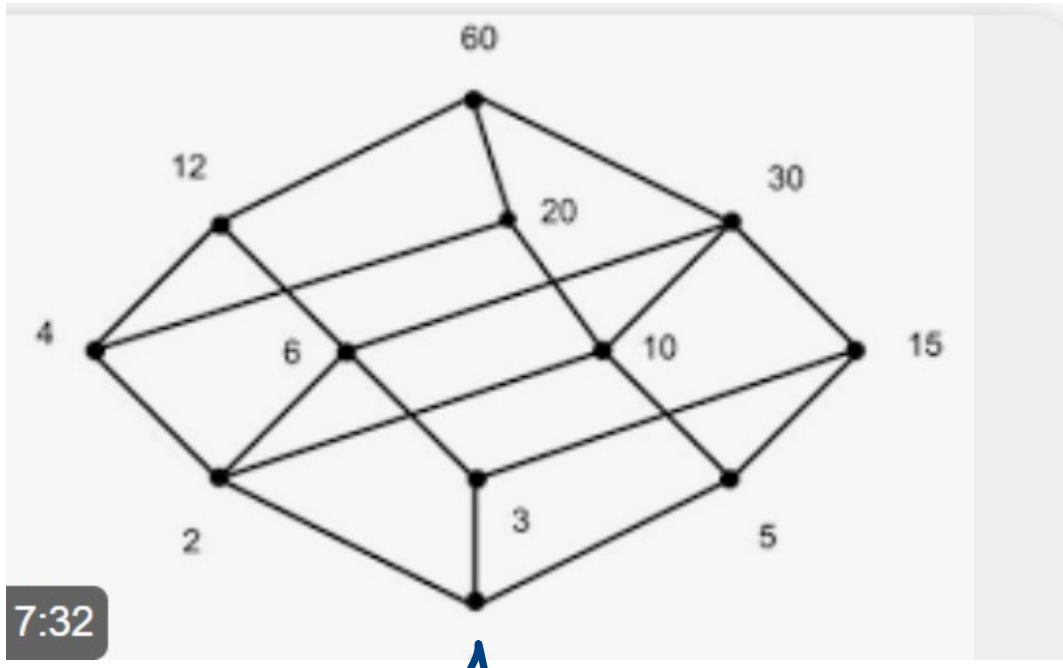

48
maximal
6
8
3
2
7
maximal
1
minimal
minim

6.3.3.7 divide a

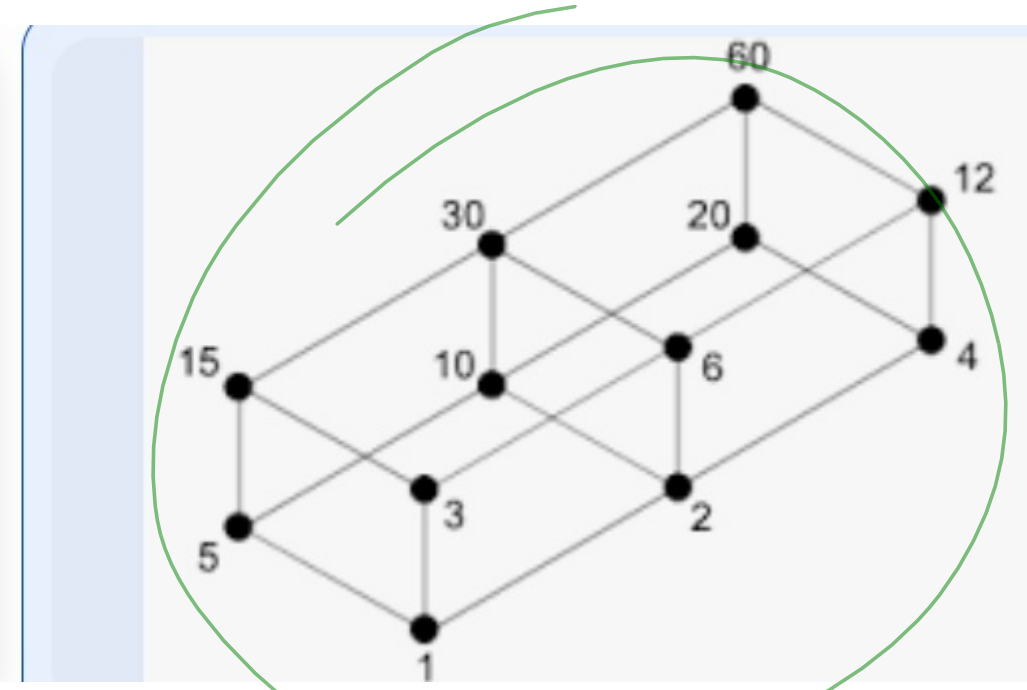
1 2 4 5 10 12 20 25 60 300



1



1



2. Determina se a relación R no conxunto dos números naturais é reflexiva, simétrica, antisimétrica e/ou transitiva, onde $(x, y) \in R$ se, e só se,

a) $x = y^2$ c) $2|(x - y)$ e) $x = 2$

b) $x \geq 3, y \geq 2$ d) $x^2 y$ é par f) $x + 3 \geq 3y + 2$

- *Reflexiva:* $(x, x) \in R$ *para todo* $x \in \mathbb{N}$
- *Simétrica: Se* $(x, y) \in R$, *entón* $(y, x) \in R$
- *Antisimétrica: Se* $(x, y) \in R$ *e* $(y, x) \in R$, *entón* $x = y$
- *Transitiva: Se* $(x, y) \in R$ *e* $(y, z) \in R$, *entón* $(x, z) \in R$

$x = y^2$

$(x,x) \rightarrow x = x^2 \rightarrow 1 = 1^2,\ 2 \neq 2^2 \rightarrow$ no reflexiva

$x = x^2 \Rightarrow x^2 - x = 0 \Rightarrow x(x-1) = 0 \begin{cases} x = 0 \\ x = 1 \end{cases}$

$\forall x \neq 0, x \neq 1$ no reflexiva

$xRy \Leftrightarrow yRx \rightarrow \begin{cases} x = y^2 \\ y = x^2 \end{cases} \rightarrow y = y^4 \begin{cases} y = 0 \\ y = 1 \end{cases}$

no simétrica

$$\left.\begin{array}{l} x = y^2 \\ y = x^2 \end{array}\right\} \Leftrightarrow \boxed{x = y}$$

$$\left.\begin{array}{l} xRy \\ yRx \end{array}\right\} \Rightarrow x = y$$

$$y = y^4 \Rightarrow \begin{array}{l} y = 0 \\ y = 1 \end{array}$$

$$x = 1^2 \Rightarrow x = 1$$

$$x = 0 \Rightarrow x = 0$$

$$\left.\begin{array}{l} x = y^2 \\ y = z^2 \end{array}\right\} \Rightarrow x = z^4 \quad \text{no transitive}$$

$$\begin{array}{l} xRy \\ yRz \end{array} \Rightarrow xRz$$

$x \geq 3$
$y \geq 2$

xRx ?

$(5,5)$

$x \geq 3$
$x \geq 2$ $\boxed{\text{Si}}$

$x \geq 3$, $y \geq 2$

$(3,2)$ No simétrica
$(2,3)$

xRy
yRx

$y \geq 3$, $x \geq 2$

$4,3$
$3,4$

xRy
yRz $\mapsto$ xRz

$x \geq 3$ $y \geq 2$

$(4,3)$
$(3,5)$ $\mapsto$ $(4,3)$

$(4,3)$
$(3,5)$ $\longrightarrow$ $(4,5)$

- $x-y$ par

$x-x=0$ par reflexiva

$xRy \Rightarrow x-y$ par

$x-y=2n$

$-x+y=-2n \Rightarrow y-x=-2n$ par

simétrica

transitiva

$x-y$ par $\Rightarrow x-y=2n_1 \Rightarrow x-2n_2-z=2n_1 \Rightarrow$

$y-z$ par $\Rightarrow y-z=2n_2$

$\Rightarrow x-z=2n_1+2n_2 \Rightarrow$

$\Rightarrow x-z=2(n_1+n_2) \Rightarrow$

$\Rightarrow x-z=2n$ par

Sí!

$(2n)^2 = 4n^2 = 2(2n^2) \Rightarrow$ par

$(2n+1)^2 = 4n^2 + 4n + 1 = 2(2n^2 + 2n) + 1 \rightarrow$ impar

$(2n)(2n+1) = 4n^2 + 2n = 2(2n^2 + n)$ par

$x^2 + y$ ~~impar~~ PAR

x impar $\rightarrow x^2$ impar $\rightarrow y$ par

x par $\Rightarrow x^2$ par $\Rightarrow y <$ par / impar

x^2 (X) No reflexiva

x impar $\rightarrow x^2$ impar $\rightarrow x^2 x$ impar

$(3,3)$ $3^2 \cdot 3 = 27$ impar

$xRy \Rightarrow x^2 y$ par

$x^2 y =$ par

$y^2 x =$ par

x par $<$ par / impar

x impar e y par

par impar $\rightarrow$ impar

(impar, par)

(par, impar)

xRy

yRx simétrica

(par, *)

(*, par)

xRy $\quad$ $x^2 y$ par

yRz $\quad$ $y^2 z$ par

$\boxed{x \text{ par}} \rightarrow (x,y)$ $\quad$ $(x,z) \rightarrow (\text{par}, *)$

(y,z)

$\otimes R_y$

$\otimes R_z$ $\rightarrow$ $\otimes R_z$

par

$x^2 z$ par

x impar y par

yRz

1Ry

yR3 $\longmapsto$ 1 ~~R~~ 3

(1, 4) $\rightarrow$ ~~(1, 3)~~

(4, 3)

xRy

yRz

Im^2 par $\rightarrow$ par

par^2 impar $\rightarrow$ par

Im^2 Impar $\rightarrow$ ~~impar~~

$x+3 \geq 3y+2$

$x+3 \geq 3x+2 \Rightarrow$

$\Rightarrow 3x+2-x-3 \leq 0 \Rightarrow 2x-1 \leq 0 \Rightarrow$

$\Rightarrow x \leq \frac{1}{2}$

$1+3 \not\geq 3+2 \rightarrow x=1$ no es reflexivo

$xRy \rightarrow (3,1)$ $6 \geq 3+2$ ✓

$(5,1)$

$5+3 \geq 3+2 \rightarrow 8 \geq 5$ $(1,-1)$

No transitive

$(1,5)$ $1+3 \not\geq 15+2$

$(4,4)$ $4+3 \not\geq 12+2$

$9z-1 \geq 3z+2 \Rightarrow$

$\rightarrow 6z \geq 1 \rightarrow$

$\rightarrow z \geq \frac{1}{6}$

$x+3 \geq 3y+2 = 9z-1$

$y+3 \geq 3z+2 \Rightarrow 3y+9 \geq 9z+6 \rightarrow$

$\rightarrow 3y+2 \geq 9z-1$